Yembuluri H N Venkata Satya Narasimha

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Summary

Final-year B.Tech student in Artificial Intelligence and Machine Learning with strong foundations in data structures, algorithms, computer organization, and system-level programming. Proficient in Python, C/C++, and analytical problem solving, with hands-on experience in designing and optimizing machine learning and computer vision solutions. Skilled in working with large-scale datasets, improving model performance, and building scalable, performance-driven engineering systems. Passionate about developing intelligent software solutions, understanding hardware–software interaction, and solving complex real-world technical challenges through structured and data-driven approaches.

Education

Sagi Rama Krishnam Raju Engineering College *Aug 2022 - Apr 2026*
B.Tech in Artificial Intelligence and Machine Learning

- **GPA**: 8.69/10
- **Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, NLP, Computer Vision

Internships

Altair Data Science Master Virtual Internship *Jan 2026 - Mar 2026*
EduSkills (AICTE Virtual Internship Program) – Cohort 15

- Worked on end-to-end machine learning workflows including data preprocessing, feature engineering, model training, and performance evaluation using real-world datasets.
- Designed structured data processing pipelines and experimented with multiple models to improve prediction reliability and analytical insights.
- Gained practical exposure to enterprise-level AI use cases, data-driven decision making, and scalable analytics solution development.
- Strengthened proficiency in Python-based data science tools and model validation techniques through continuous hands-on assessments.

AI-ML Virtual Intern (Google for Developers Program) *Oct 2025 - Dec 2025*
EduSkills Academy – Cohort 14

- Built and evaluated supervised learning models for classification and regression tasks using Python and Scikit-learn on structured datasets.
- Implemented data preprocessing, feature selection, and model optimization techniques to enhance predictive performance.
- Analyzed model results using evaluation metrics such as accuracy, precision, recall, and F1-score to support data-driven insights.
- Explored practical applications of machine learning in predictive analytics and intelligent automation scenarios.

Technical Skills

Programming Languages: Python, C, C++, Java, SQL

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming (OOP), Computer Organization & Architecture, Operating Systems Fundamentals, Analytical Problem Solving

Machine Learning & Computer Vision: Scikit-learn, TensorFlow, Keras, Convolutional Neural Networks (CNNs), Supervised Learning, Feature Engineering, Model Tuning, Model Validation

Data Analysis & Processing: Pandas, NumPy, Matplotlib, Exploratory Data Analysis (EDA), Statistical Analysis, Performance Metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC)

Tools & Platforms: Git, GitHub, MySQL, Firebase, Linux Basics, RESTful APIs

Projects

German Traffic Sign Recognition using Deep Learning

- Built a Convolutional Neural Network (CNN) model to classify **43 German traffic sign categories** using the GTSRB dataset with **50,000+ labeled images**.
- Improved model generalization and inference performance through data augmentation, normalization, and systematic hyperparameter tuning.
- Evaluated classification performance using confusion matrix visualization and metrics including precision, recall, and F1-score.
- **Technologies Used:** Python, TensorFlow, Keras, OpenCV, NumPy, Matplotlib

Machine Learning-Based Fraud Detection System

- Developed a supervised learning pipeline to detect fraudulent financial transactions from highly imbalanced datasets.
- Applied feature engineering and **SMOTE-based resampling** to enhance minority class detection sensitivity and overall model robustness.
- Optimized predictive performance using ROC-AUC analysis and precision–recall trade-off evaluation.
- **Technologies Used:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, SQL

Community Connect Hub – Local Community Service Platform

- Developed a responsive frontend web platform enabling users to explore local community services and events through structured and user-friendly interface components.
- Implemented secure user authentication and session handling using **Firebase Authentication**.
- Designed modular UI workflows to enhance navigation efficiency and overall user engagement across service discovery and event interaction features.
- **Live Demo:** Community Connect Hub Project
- **Technologies Used:** Angular / HTML / CSS / JavaScript, Firebase Authentication

Achievements

- Secured **1st Prize in Art Fusion (Nov 2024)** organized by SRKR Event Organizing Cell for creative technical presentation and structured exhibit design.
- Achieved **Semifinalist position in SRKR Orator Championship**, demonstrating strong public speaking, analytical articulation, and technical communication skills.
- Played a key role in **organizing large-scale 24-hour hackathons and institutional events**, contributing to planning, coordination, and smooth technical execution.
- Presented machine learning project outcomes to peer groups and mentors, enhancing **model interpretability awareness and collaborative technical learning**.